CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

7.5 Polarisation

Syllabus Section: 7. Waves **Paper Type:** Structured Questions (Paper 2) **Total:** 7 Questions (58 Marks)

Question 1 Easy [8 marks]

Part (a) [2 marks]

Transverse waves such as light can be polarised.

Define the term *polarised*.

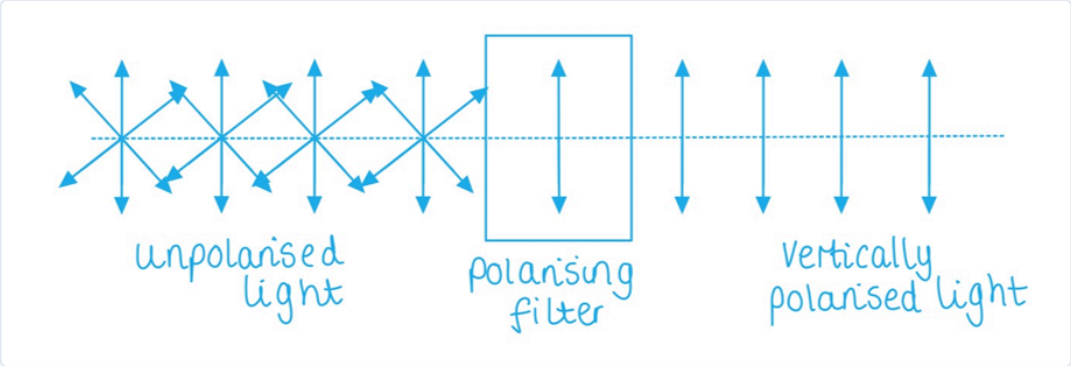
Mark Scheme (2 marks):

Polarised waves are waves which...

- Have their oscillations / vibration restricted; *[1 mark]*
- To one plane / direction; *[1 mark]*

[Total: 2 marks]

Below is a diagram of an unpolarised transverse waves passing through a polarising filter. The transmitted light is polarised in the vertical plane.



Part (b) [3 marks]

Explain why longitudinal waves cannot be polarised by a polarising filter.

Mark Scheme (3 marks):

Longitudinal waves cannot be polarised because...

- Their oscillations are parallel to the direction of wave travel; *[1 mark]*
- Polarising filters restrict oscillations that are perpendicular to the direction of wave travel; *[1 mark]*
- Therefore, (in longitudinal waves) there are no oscillations that can be restricted by a polarising filter; *[1 mark]*

[Total: 3 marks]

Longitudinal waves cannot be polarised at all. This is one of the characteristics of longitudinal waves.

Part (c) [2 marks]

Fig. 1.1 shows unpolarised light directed towards a polarising filter A.

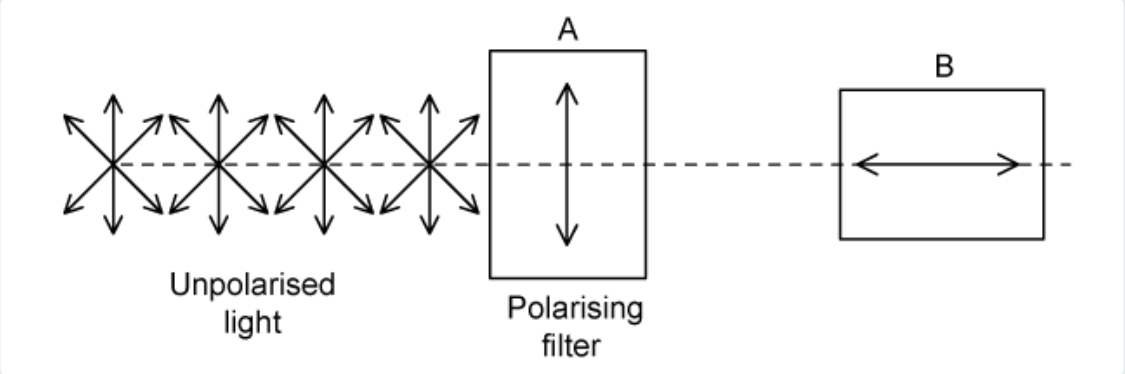


Fig. 1.1

Polarising filter B is placed directly after A and is identical except rotated by 90°.

State and explain what happens to the light after it has passed through polarising filter B.

Mark Scheme (2 marks):

After passing through polarising filter B:

- There is no light / the light doesn't transmit through; *[1 mark]*

Because:

- The light is vertically polarised after (polarising filter) A

OR

- Only horizontally polarised light can transmit through (polarising filter) B; *[1 mark]*

[Total: 2 marks]

Part (d) [1 mark]

In Fig. 1.1 the incident light has an intensity of I_0.

State the intensity of the polarised light exiting polarising filter A.

Mark Scheme (1 marks):

The intensity of the polarising light exiting filter A is...

- Half of I_0 **OR** $0.5I_0$ **OR** $\frac{I_0}{2}$ *[1 mark]*

[Total: 1 mark]

> *When unpolarised light passes through the first polariser, half of the intensity of the wave is always lost. This is the* ***half rule****.*

Question 2 Easy [9 marks]

Part (a) [2 marks]

State **two** ways that light can be polarised other than by a polarising filter.

Mark Scheme (2 marks):

Ways that light can be polarised:

*Any **two** from:*

- Reflection; *[1 mark]*
- Refraction; *[1 mark]*
- Scattering; *[1 mark]*

[Total: 2 marks]

Part (b) [2 marks]

Fig. 1.1 shows unpolarised light of intensity 64 W m^{-2} incident on polariser A.

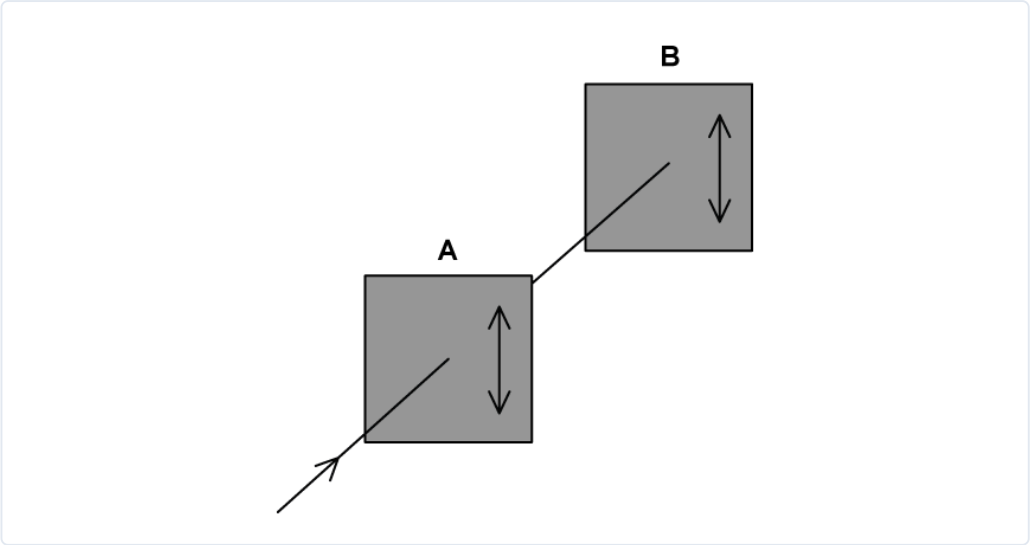


Fig. 1.1

Determine the intensity of the light leaving the analyser B.

Mark Scheme (2 marks):

To determine the intensity of the light leaving analyser B:

List the known quantities:

- Intensity of incident light, I_0 = 64 W m^{-2}

Determine the intensity of the polarised light, I_1 incident on analyser B:

- When unpolarised light passes through a polariser, the intensity of the light falls by half
- Therefore, $I_1 = \frac{I_0}{2} = \frac{64}{2} = 32 \text{ W m}^{-2}$ *[1 mark]*

Determine the intensity, I, of the light leaving the analyser:

- The analyser is at the same angle as the polariser
- Therefore, all the polarised light will pass through the analyser
- $I = I_1 = 32 \text{ W m}^{-2}$ *[1 mark]*

[Total: 2 marks]

Always remember to apply the half rule if the question includes the light passing through the polariser before the analyser.

Part (c) [3 marks]

Fig. 1.2 shows that analyser B has been rotated by 35° with respect to the polariser A.

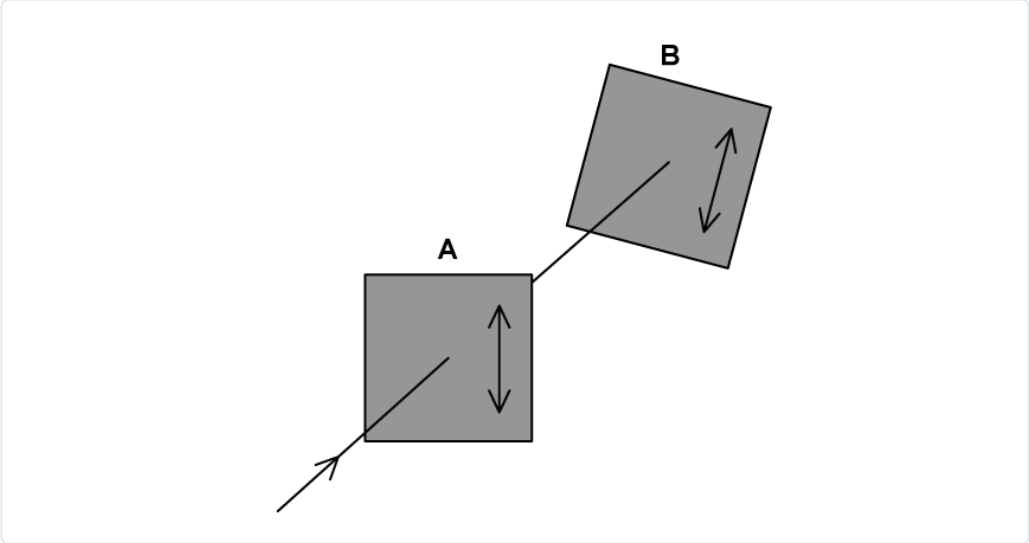


Fig. 1.2

Determine the intensity of the light leaving analyser B.

Mark Scheme (3 marks):

To determine the intensity of the light leaving analyser B:

List the known quantities:

- Intensity of incident light, I_0 = 64 W m^{-2}
- Intensity of polarised light, I_1 = 32 W m^{-2} (from part (b))
- Angle of analyser, θ = 35°

Determine the intensity, I, of the light leaving the analyser:

- Recall Malus' Law, $I = I_1 \cos^2 \theta$ *[1 mark]*
- $I = 32 \times \cos^2 35$ *[1 mark]*
- $I = 21 \text{ W m}^{-2}$ 2 s.f. *[1 mark]*

[Total: 3 marks]

Remember that Malus' law considers the light incident on the analyser as I_0 in $I = I_0 cos^2 \theta$, where I_0 is the maximum intensity that the light passing through the analyser could possibly be. If the question considers the light passing the through the polariser as well, then the original intensity of the unpolarised light will be marked as I_0 and polarised light incident on the analyser can be denoted I_1

In terms of typing $\cos^2 \theta$ into your calculator, you sq uare the answer to $\cos\theta$.

Part (d) [2 marks]

Fig. 1.3 shows that analyser B has been rotated to 90° with respect to polariser A.

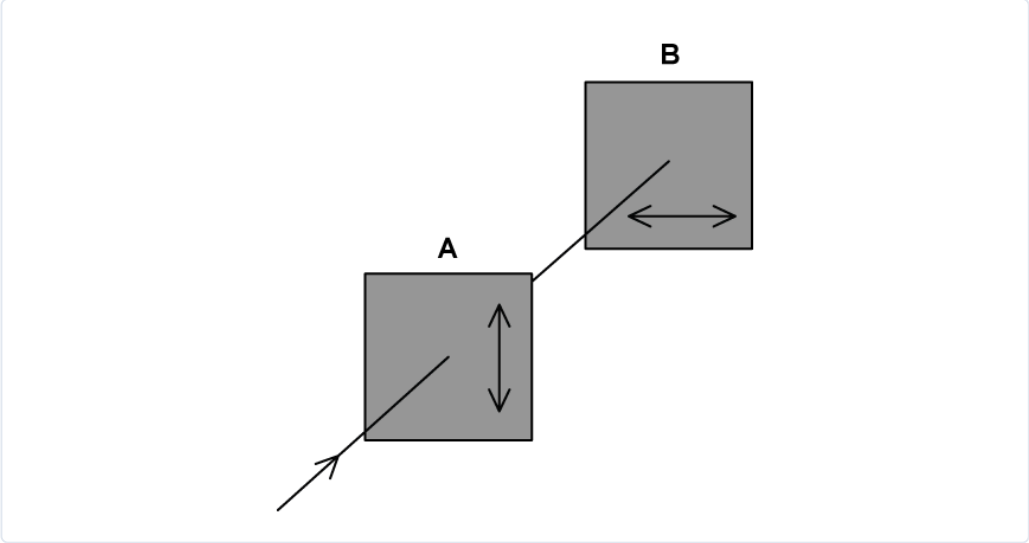


Fig. 1.3

Show that there is no light passing through analyser B.

Mark Scheme (2 marks):

To show that there is no light passing through analyser B:

List the known quantities:

- Maximum intensity = Intensity of polarised light, I_1 = 32 W m^{-2}
- Angle of analyser, θ = 90°

Use Malus' Law:

- $I = I_1 \cos^2 \theta$

- $\cos 90 = 0$ *[1 mark]*
- Therefore, $I = 32 \times \cos^2 90 = 0$ W m^{-2} *[1 mark]*

[Total: 2 marks]

Remember, again, that Malus' law considers the light incident on the analyser as I_0 in $I = I_0 \cos^2 \theta$, where I_0 is the maximum intensity that the light passing through the analyser could possibly be. If the question considers the light passing the through the polariser as well, then the original intensity of the unpolarised light will be marked as I_0 and polarised light incident on the analyser can be denoted I_1 (this is then the maximum intensity that the light passing through the analyser could possibly be).

Question 3 Medium [5 marks]

Part (a) [1 mark]

A beam of vertically polarised monochromatic light is incident on a polarising filter, as shown in Fig. 5.1.

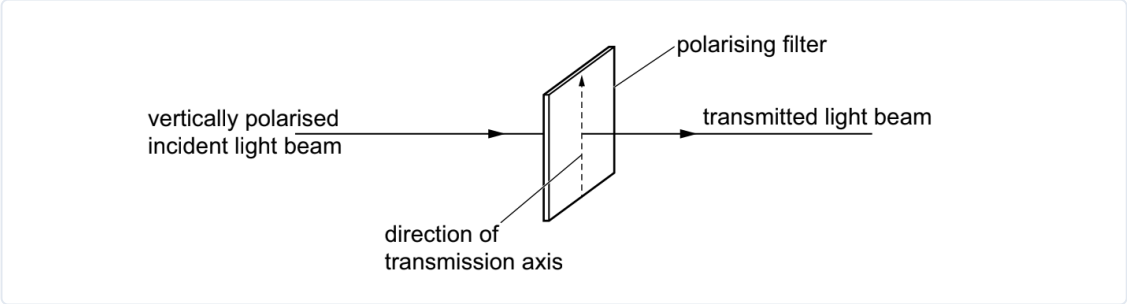


Fig. 5.1

The transmission axis of the filter is initially vertical and the transmitted light beam has the same intensity as the incident light beam.

The filter may be rotated about the direction of the light beam to change the angle of the transmission axis against the vertical.

State one angle of the transmission axis to the vertical that results in no transmitted light beam.

angle = ... °

Mark Scheme (1 marks):

No light will be transmitted through the polarising filter if the transmission axis is rotated to be:

- 90° (to the vertical); *[1 mark]*

[Total: 1 mark]

Part (b) [4 marks]

The filter is now positioned with its transmission axis at angle θ to the vertical, as shown in Fig. 5.2.

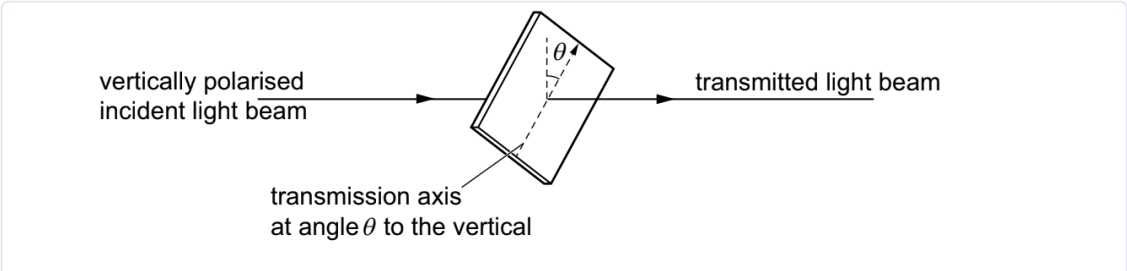


Fig. 5.2

The ratio $\frac{\text{intensity of transmitted light}}{\text{intensity of incident light}}$ is equal to 0.75.

(i) Calculate angle θ.

θ = ... ° [2]

(ii) Calculate the ratio

$$\frac{\text{amplitude of transmitted light}}{\text{amplitude of incident light}}.$$

ratio = ... [2]

Mark Scheme (4 marks):

(i) Calculate angle θ:

List the known quantities:

- $\frac{\text{intensity of transmitted light}}{\text{intensity of incident light}} = \frac{I}{I_0} = 0.75$

Use Malus's Law to obtain a value for the ratio $\frac{I}{I_0}$:

- $I = I_0 \cos^2 \theta$
- $\frac{I}{I_0} = \cos^2 \theta = 0.75$ *[1 mark]*

Calculate angle θ:

- $\cos \theta = \sqrt{0.75}$
- $\theta = \cos^{-1} \sqrt{0.75} = 30°$ *[1 mark]*

(ii) Calculate the ratio $\frac{A}{A_0}$:

- $I \propto A^2$ *[1 mark]*
- $\frac{I}{I_0} \propto \left(\frac{A}{A_0}\right)^2$
- $0.75 \propto \left(\frac{A}{A_0}\right)^2$
- $0.75^{\frac{1}{2}} \propto \frac{A}{A_0}$
- Therefore, $\frac{A}{A_0} = 0.866 = 0.87$ $2\ s.f.$ *[1 mark]*

[Total: 4 marks]

*Linking **Malus's Law** to the **ratio given** is the key to answering part (i).*

Question 4 Medium [6 marks]

Part (a) [3 marks]

Unpolarised light is passed through a pair of polarising sunglasses as shown in Fig. 1.1.

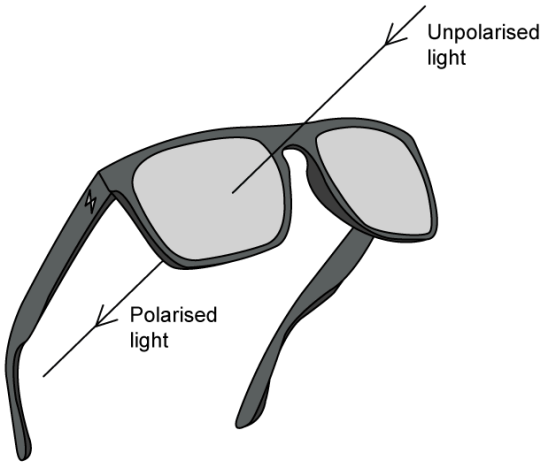


Fig.1.1

The light becomes plane polarised.

Explain the difference between unpolarised and plane polarised light.

Mark Scheme (3 marks):

Explain the difference between unpolarised and plane polarised light:

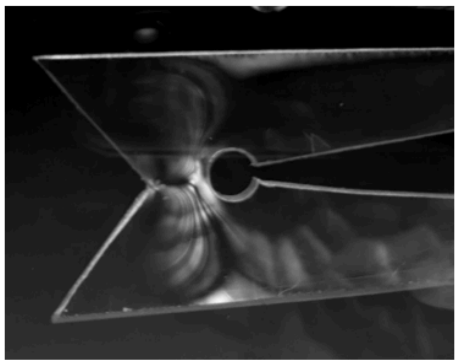
- Unpolarised light vibrates / oscillates in all planes; *[1 mark]*
- Plane polarised light vibrates / oscillates in one plane; *[1 mark]*
- (Planes of oscillation are) perpendicular to the direction of wave travel; *[1 mark]*

[Total: 3 marks]

Part (b) [3 marks]

Polarisation can be used to identify when some materials are under stress. Placing the material under stress causes the plane of polarisation of the light passing through the material to be rotated. The greater the stress, the greater the rotation of the plane of polarisation of the light.

A piece of transparent material under stress is positioned between two polarising filters. There is an angle of 90° between the planes of polarisation of the two polarising filters.



Explain how this photograph can be used to identify areas of the material with different amounts of stress.

Mark Scheme (3 marks):

Explain how this photograph can be used to identify areas of the material with different amounts of stress:

- (Polarising) filters at 90° to each other do not allow light to pass through; *[1 mark]*
- Rotation of plane of polarisation (due to stress) allows light to pass; *[1 mark]*
- Darker areas represent less stress **OR** brighter areas represent greater stress; *[1 mark]*

[Total: 3 marks]

Question 5 Medium [7 marks]

Part (a) [2 marks]

Fig. 1.1. shows two ways in which a wave can be made to travel along a slinky spring.

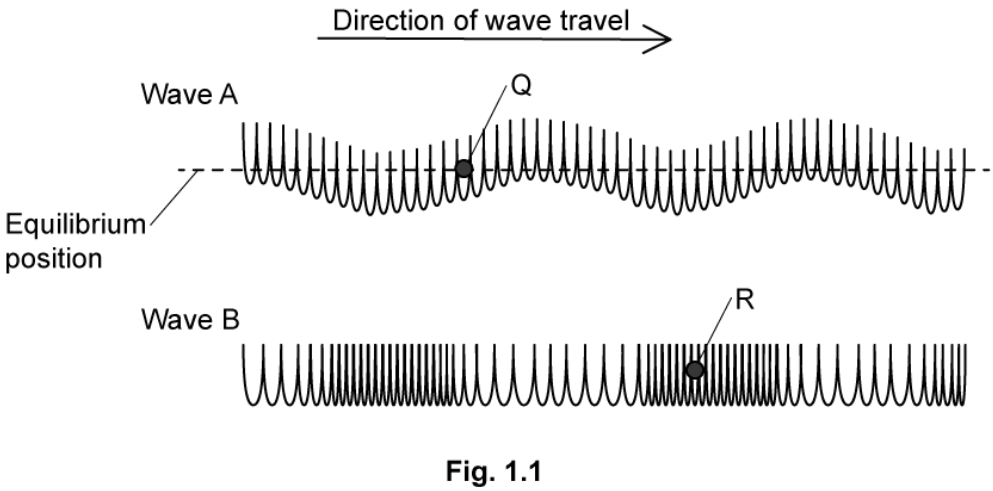


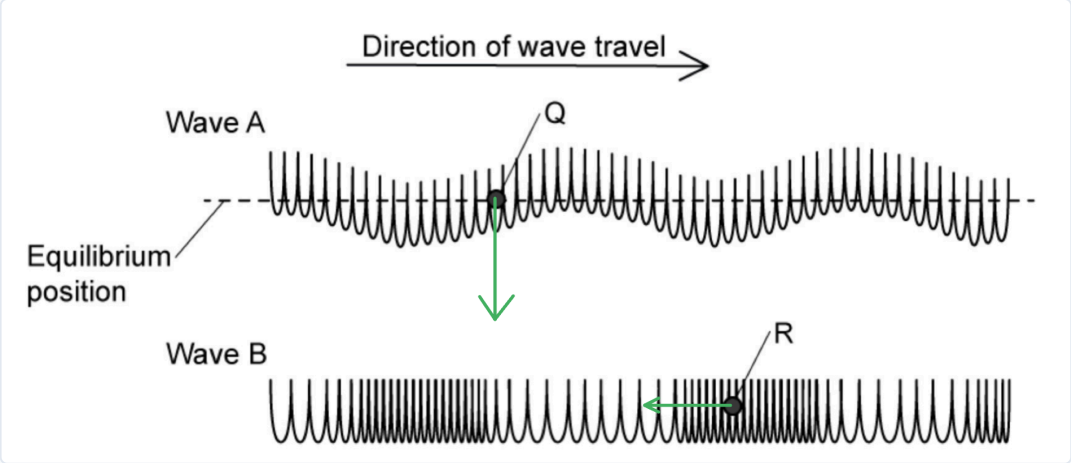
Fig. 1.1

Use arrows to show how points Q and R will move next as the two waves move to the right.

Mark Scheme (2 marks):

Use arrows to show how points Q and R will move next as the two waves move to the right:

Correctly drawn arrows look like:



- Q moving downwards; *[1 mark]*
- R moving to the left; *[1 mark]*

[Total: 2 marks]

You may have thought that R will follow the direction of the wave. However, it is to the left of the centre of a compression. Therefore as the compression expands to a rarefaction, R will be pushed backwards compared to the

direction of travel of the wave.

Part (b) [3 marks]

Use the diagram in Fig. 1.1.

(i) State which wave in Fig. 1.1 can be used to model an electromagnetic wave.

[1]

(ii) Hence explain why it is important to correctly align a radio antenna in order to receive the strongest signal.

[2]

Mark Scheme (3 marks):

Use the diagram in Fig. 1.1.

(i) State which wave can be used to model an electromagnetic wave:

- Wave A can be used to model electromagnetic waves; *[1 mark]*

(ii) Hence explain why it is important to correctly align a radio antenna in order to receive the strongest signal.

- Transmitted radio waves are polarised; *[1 mark]*
- Antennae must be aligned to the same angle / plane of polarisation; *[1 mark]*

[Total: 3 marks]

Part (c) [2 marks]

Light from a filament lamp is viewed through two polarising filters **A** and **B** as shown in Fig. 1.2.

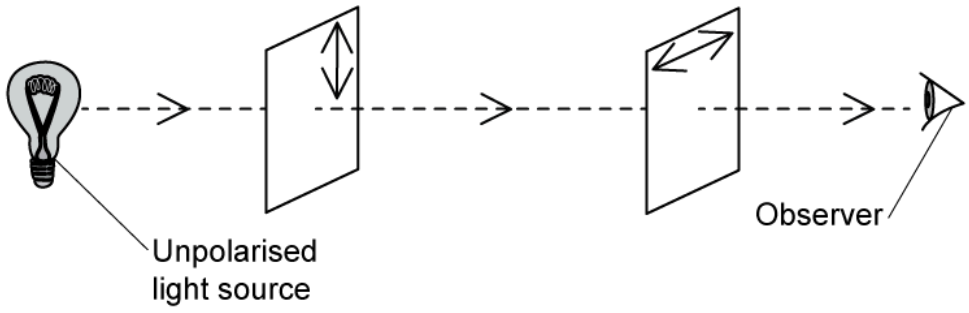


Fig. 1.2

The direction of plane polarisation is indicated on each filter using a double-headed arrow.

Explain why the observer cannot see light from the filament lamp.

Mark Scheme (2 marks):

Explain why the observer cannot see light from the filament lamp:

- Light passing through polarising filter **A** is vertically polarised; *[1 mark]*
- Only light in the horizontal plane / horizontally polarised light will pass through polarising filter **B** but after being vertically polarised no light can pass through; *[1 mark]*

[Total: 2 marks]

Question 6 Hard [12 marks]

Part (a) [4 marks]

A transverse wave passes straight through a polariser. A and B are two points on the wave a distance of $\frac{3}{2}\lambda$ apart. The displacement of the wave at the point when A passes through the polariser is at its maximum.

Sketch a possible graph to show the displacement of the wave against phase angle θ as it passes through the polariser, and label points A and B.

Mark Scheme (4 marks):

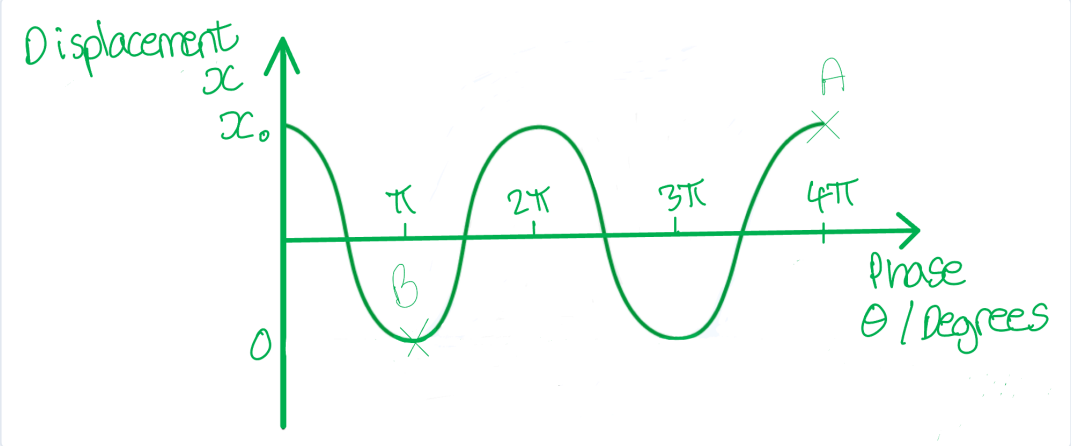
Determine the possible displacement angles when A passes through the polariser:

- When A passes through the polariser, the amplitude is maximum, so the displacement angle is 0, 180 or 360 degrees; *[1 mark]*

Determine the possible displacement angles when B passes through the polariser:

- A and B are $\frac{3}{2}\lambda$ apart / out of phase by $\frac{3}{2}\lambda$
- So, when B passes through the polariser, the amplitude is a minimum and the displacement angle is 90 or 270 degrees; *[1 mark]*

A graph to show wave amplitude against displacement angle / phase should look like this:



- Sinusoidal graph drawn with a maximum displacement (amplitude) x_0 at 0, 180 and 360 degrees; *[1 mark]*
- B drawn at 90 degrees and A drawn $\frac{3}{2}\lambda$ in front at 360 degrees; *[1 mark]*

[Total: 4 marks]

Don't get confused with the graph of intensity and transmission angle. This is what is normally related to a polariser. The question is asking you to sketch a graph showing the phase difference of a wave passing straight through a polariser, so nothing to do with polarisation at all.

Remember that the wave is travelling from left to right, so part A will be in front of part B. There is only one combination of correct angles that will fit on this graph.

Part (b) [3 marks]

Electromagnetic waves are transverse waves that carry energy through the vacuum of space as progressive transverse waves.

Discuss the motion of an electromagnetic wave with reference to the appropriate fields.

Mark Scheme (3 marks):

Describe the motion of the fields which comprise an electromagnetic wave:

- Electromagnetic waves are made of electric and magnetic fields; *[1 mark]*
- These fields oscillate at 90 degrees / perpendicular to each other; *[1 mark]*

Describe the direction of energy transfer:

- (Because electromagnetic waves are transverse) energy transfer is 90 degrees / perpendicular to the direction of (field) oscillations; *[1 mark]*

[Total: 3 marks]

Part (c) [3 marks]

Plane-polarised electromagnetic waves are incident on an aerial positioned to give a maximum response.

High winds cause the aerial to rotate anti-clockwise about the plane of the incident wave until it makes an angle of 40° to the plane of wave oscillation.

Calculate the percentage reduction in the amplitude of the signal received by the aerial following the high winds.

Mark Scheme (3 marks):

List the known quantities:

- Change in direction of the aerial = 40°

Find an expression linking the initial and final amplitude of the waves following polarisation:

- Malus's law: $I = I_0 \cos^2 \theta$
- Since $I \propto A^2$:
  - $\frac{I}{I_0} = \frac{A}{A_0}^2 = \cos^2 \theta$
  - $\frac{A}{A_0} = \cos \theta$
- Therefore, $A = A_0 \cos \theta$ *[1 mark]*

Find an expression for the percentage reduction:

- $A = A_0 \cos 40$
- Percentage reduction = $\frac{\text{initial amplitude - final amplitude}}{\text{initial amplitude}} \times 100\%$

- Percentage reduction = $\frac{A_0 - A_0 \cos 40}{A_0} \times 100\%$ *[1 mark]*

Evaluate to find percentage reduction:

- $\frac{A_0 - A_0 \cos 40}{A_0} \times 100\% = 1 - \cos 40 \times 100\% = 23.4\%$ *[1 mark]*

[Total: 3 marks]

Part (d) **[2 marks]**

The high winds cause permanent damage to the aerial. After the aerial is repaired, it remains at an angle of 15° to the plane of wave oscillation and is found to oscillate with an amplitude of 54.0 µm. The intensity of the electromagnetic waves received at this angle is measured to be 0.056 mW m^{-2}.

When the aerial is positioned to give a maximum response, it is found to oscillate with an amplitude of 55.9 µm.

Determine the intensity of the incident electromagnetic waves.

Mark Scheme (2 marks):

List the known quantities:

- Maximum vibration amplitude, A_0 = 55.9 µm, when $\theta = 0°$
- Measured vibration amplitude, A = 54.0 µm, when $\theta = 15°$
- $A = A_0 \cos \theta$
- $I \propto A^2$

Determine the ratio between the intensities:

- The maximum response is achieved when the aerial is perfectly aligned with the plane of wave oscillation, where $\theta = 0°$ and $I = I_0$
- Since intensity is proportional to the square of amplitude:
  - $I_0 \propto A_0^2$
  - $\frac{I}{I_0} = \frac{A}{A_0}^2$
- $\frac{I}{I_0} = \frac{54.0}{55.9}^2 = 0.933$ *[1 mark]*

Determine the incident intensity of the waves:

- $I_0 = \frac{I}{0.933}$
- $I_0 = \frac{0.056}{0.933} = 0.060$ mW m^{-2} *[1 mark]*

[Total: 2 marks]

Question 7 Hard [11 marks]

Part (a) [2 marks]

When electromagnetic waves are reflected from a shiny surface, such as a road sign, they often become polarised.

Suggest how to determine experimentally if visible light reflected from a road sign is polarised.

Mark Scheme (2 marks):

State the use of a polaroid filter:

- Hold a polaroid filter up to the reflected light; *[1 mark]*

Describe the effect of rotating the polaroid filter:

- Rotating the polaroid filter will determine if the light is polarised

  AND

- If the light is polarised (along a certain plane) the polaroid filter will not transmit it at some angle; *[1 mark]*

[Total: 3 marks]

Generally speaking, it's really good practice to specify angles of rotation in polaroid filter questions. For example, if the intensity of light transmitted behind two polaroid filters is 100% when the two filters are aligned at an angle θ, then the light intensity will fall to 0% when this angle is (θ + 90°) and then back up to 100% at (θ + 180°) and back to 0% at (θ + 270°) and then finally back to 100% at (θ + 360°) – yes really, this much detail! You will usually secure full credit for specific descriptions through a full rotation.

However, in this question, it is sufficient to refer to 'some angle θ' because you are only required to determine if the light is polarised or not. A full description of the effects on the intensity of transmitted light through a full rotation is not required.

Part (b) [3 marks]

Three polaroid filters P_1, P_2 and P_3 are aligned as shown in Fig. 1.1.

Unpolarised light is incident on P_1 and subsequently passes through each of the three polaroid filters. P_1 and P_2 are fixed, with P_2 at 20° to P_1, but P_3 can be rotated to any angle θ to that of P_1.

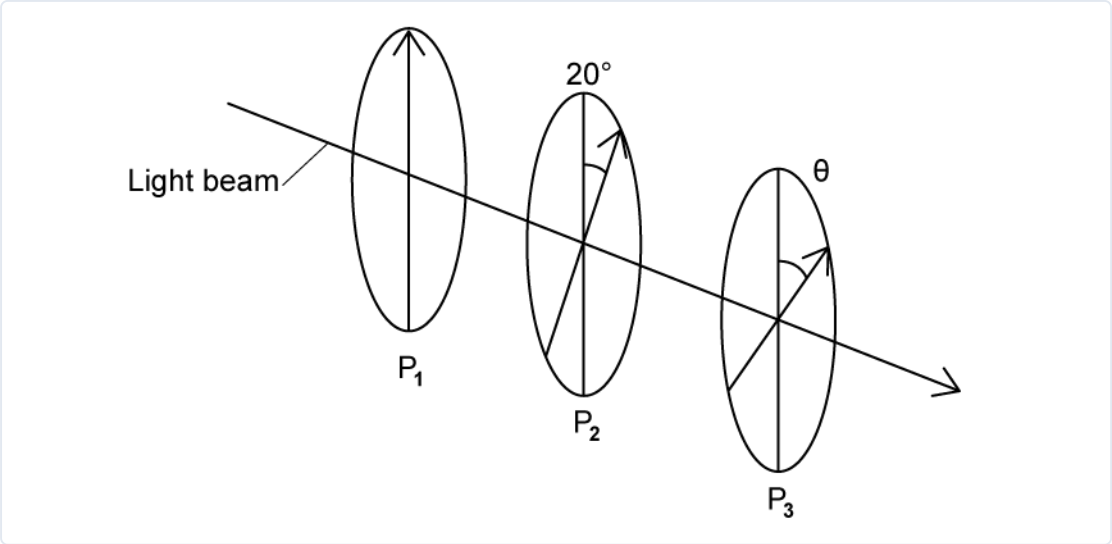


Fig. 1.1

Determine the angles of θ at which minima and maxima of emergent light intensity occur.

Mark Scheme (3 marks):

State the angles of θ with respect to the first polaroid filter P_1:

- Polarised light, parallel to the orientation of P_2, is incident on P_3
- This means that, when $\theta = 20^{\circ}$ emergent light is at maximum intensity (P_3 is parallel to the alignment of P_2); *[1 mark]*

Determine the angles of minimum and maximum intensity:

- $\theta = (20^{\circ} + 90^{\circ}) = 110^{\circ}$ is the first angle of minimum intensity; *[1 mark]*
- $\theta = (20^{\circ} + 180^{\circ}) = 200^{\circ}$ is the first angle of maximum intensity

  AND

  $\theta = (20^{\circ} + 270^{\circ}) = 290^{\circ}$ is the second angle of minimum intensity; *[1 mark]*

[Total: 3 marks]

Part (c) **[2 marks]**

Table 1.1 lists different waves, whether they are polarised or unpolarised and a reason.

State, by completing the spaces in the table whether the waves listed are polarised or unpolarised, and give a reason for your answer.

Table 1.1

Wave	Polarised or Unpolarised	Reason
Light from the sun		
Compression waves caused by an earthquake	Unpolarised	Longitudinal waves cannot be polarised
Electromagnetic waves from the sun passing through a sunglass lens		
Ultra-sonic waves from an echo sounder		

Mark Scheme (2 marks):

Identify the properties of each wave and how it is generated:

Wave	Polarised or Unpolarised	Reason
Light from the sun	*Unpolarised*	*Light in the sun is generated randomly / oscillates in all planes*
Compression waves caused by an earthquake	Unpolarised	Longitudinal waves cannot be polarised
Electromagnetic waves from the sun passing through a sunglass lens	*Polarised*	*The sunglasses act as a polariser removing one plane of oscillation of the sunlight (making it easier for people to see in the Sun)*
Ultra-sonic waves from an echo sounder	*Unpolarised*	*Sound waves cannot be polarised*

- All waves correctly identified as polarised or unpolarised; *[1 mark]*
- All reasons correct; *[1 mark]*

[Total: 2 marks]

Part (d) **[4 marks]**

Electromagnetic waves, being transverse, can also be polarised.

A light source is viewed through two pieces of polarisers, P and Q, with their axes initially at $\frac{\pi}{2}$ radians from each other, as shown in Fig. 1.1.

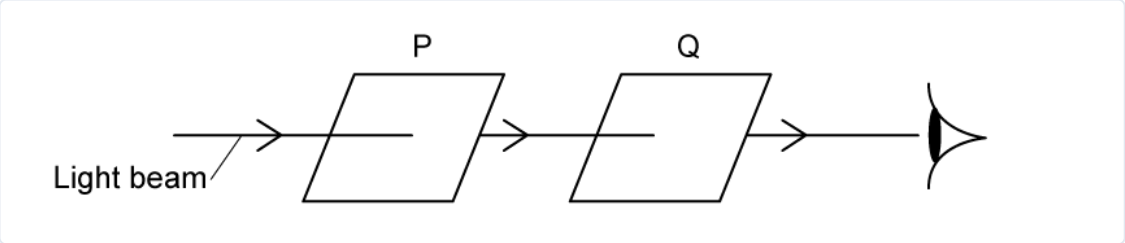
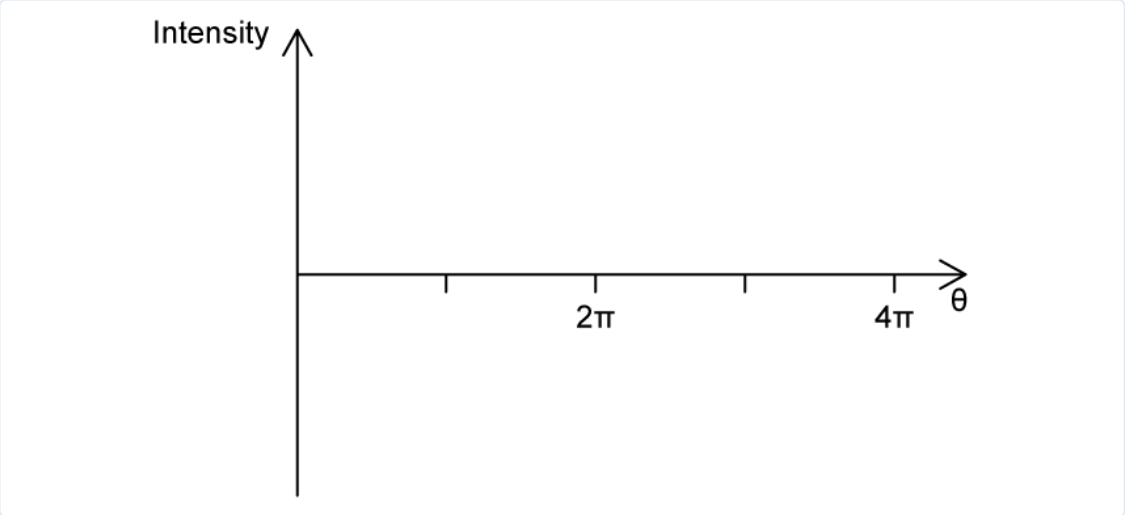


Fig. 1.1

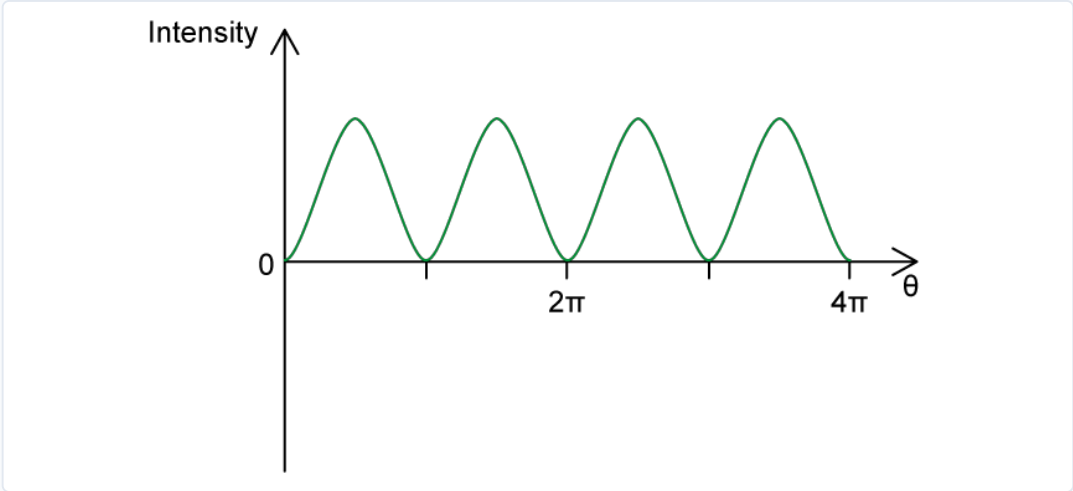
Using the axes below, sketch the variation of the intensity of light reaching the eye with angular displacement of Q with respect to P when polariser Q is rotated.



Mark Scheme (4 marks):

Draw the variation of intensity beginning from the origin:

- A sketch which would score full marks would look like this:



- Correct general shape; *[1 mark]*
- Minima at π, 2π, 3π, and 4π radians; *[1 mark]*
- Therefore, maxima at $\frac{1}{2}\pi$, $\frac{3}{2}$ π, $\frac{5}{2}$ π and $\frac{7}{2}$ π *[1 mark]*
- Constant maximum intensity; *[1 mark]*

[Total: 4 marks]

The minima occur at whole integers of π because the initial orientation is $\frac{\pi}{2}$ radians (intensity begins at zero), so after a further π radians, the intensity will be reduced to zero once more, and so on every π radians.

Examiners point out a common mistake here is to sketch a graph that looks like the following:

